

Latent Bridge Matching: Fast Artistic Transfer with Terminal SWD and Kinetic Regularization

Anonymous Authors

Abstract

Fast artistic style transfer remains difficult because style strength, content preservation, and computational efficiency often conflict. We study domain-level multi-style artistic transfer and propose a latent bridge-inspired framework that transports compact VAE latents with a style-conditioned velocity field. The method combines terminal sliced-Wasserstein distribution matching, which drives the endpoint toward the target artistic domain, with kinetic regularization, which penalizes excessive latent displacement and preserves content structure. A learnable style spatial prior and semantic cross-attention provide efficient domain-level style conditioning without per-image optimization or diffusion sampling. Under a strict 750-output protocol over five domains, the proposed model reaches a competitive raw CLIP-style score of 0.7161 while improving LPIPS-content over SaMST and substantially improving artifact-sensitive perceptual diagnostics including MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, and shallow Gram statistics. Destructive ablations show that terminal SWD is the primary style driver, kinetic regularization is essential for content preservation, and strong color matching damages the style-content trade-off. The results support a conservative but practically useful claim: latent bridge matching provides a clean, efficient, and reproducible style-content trade-off for fast multi-style artistic transfer.

1 Introduction

Artistic style transfer has long served as a testbed for separating image content from visual appearance. Classical neural style transfer showed that convolutional features and Gram statistics can synthesize compelling artistic textures [6], while feed-forward arbitrary style transfer made real-time deployment practical through normalization-based style modulation [11]. More recent attention, transformer, training-free, and diffusion-based systems have improved flexibility and style fidelity [15, 4, 9, 10, 7, 2]. However, practical multi-style artistic transfer still exposes a persistent three-way tension: strong target-style expression, faithful content preservation, and low computational cost.

This tension is especially visible in domain-level multi-style transfer. Efficient multi-style systems can support many styles and fast inference, but the output may become washed, over-smoothed, or contaminated by structured grain artifacts. Diffusion-based or training-free methods can produce strong style semantics, but may be expensive or suffer semantic drift under content-preserving evaluation. In our local strict-750 protocol, for example, StyleID obtains the highest raw CLIP-style score but exhibits severe content collapse, while SaMST obtains strong raw style and content scores but produces visible muddy and grain-like texture artifacts. These observations suggest that standard style/content metrics alone are insufficient: a competitive system must also be evaluated for perceptual quality and artifact-sensitive texture realism.

We propose a fast latent-space multi-style artistic transfer framework that models stylization as a style-conditioned latent probability flow. Instead of directly predicting RGB images, the method transports a compact VAE latent through a neural velocity field. A terminal sliced-Wasserstein distance (SWD) term aligns generated endpoints with the target artistic domain, while kinetic regularization penalizes excessive latent displacement and preserves content. A learnable style spatial prior and semantic cross-attention provide a lightweight approximation to local style transport without per-image optimization or diffusion sampling.

We call the resulting system *Latent Bridge Matching* (LBM). The name emphasizes that the method is not merely a feed-forward style injector: it matches the terminal artistic distribution while regularizing the latent path used to reach that distribution.

Our contributions are:

Table 1: Claim-evidence map used throughout the paper.

Claim	Evidence in this study
Terminal SWD drives style	Removing terminal SWD drops CLIP-style from 0.7014 to 0.6708 in destructive ablation.
Kinetic loss preserves content	Removing kinetic keeps style high but degrades CLIP-content to 0.6624 and LPIPS to 0.6375.
Ours is competitive with SaMST but cleaner	Ours is close in CLIP-style and improves MUSIQ, MANIQA, DISTs, HF-KID, FFT slope error, and Gram micro.
Raw style alone is insufficient	StyleID has the highest CLIP-style but collapses content with LPIPS 0.7497.
Balanced training is strong	The 40-run sweep finds K2 balanced default best for EC and K1 balanced default best for raw style.

- We formulate domain-level multi-style artistic transfer as a bridge-inspired latent transport problem with separate terminal style matching and path-energy content preservation.
- We introduce an efficient style spatial prior and semantic cross-attention architecture that supports fast style-conditioned latent velocity prediction.
- We provide a strict 750-output evaluation protocol with complete local comparisons to AdaIN, S2WAT, SaMST, StyleID, and our method, together with artifact-sensitive diagnostics for grain and perceptual quality.
- We conduct destructive ablations and 40-run category-weight sweeps showing that terminal SWD drives style, kinetic regularization preserves content, and naive color matching harms the style-content trade-off.

2 Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses [6]. AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics [11]. Attention-based methods such as SANet improved local style matching through style-attentional feature aggregation [15]. Recent aesthetic and pattern-aware methods, including AesPA-Net and AesFA, further emphasize pattern repeatability, aesthetic features, and high-quality arbitrary stylization [9, 10]. These methods demonstrate the value of feature statistics and local matching, but they generally do not explicitly model a style-conditioned transport path with a terminal distribution constraint.

Transformer and multi-style transfer. Transformer-based style transfer methods such as StyTr2 and S2WAT use attention mechanisms to improve content-style interaction and long-range feature modeling [4, 18]. Multi-style approaches aim to amortize many styles in a single compact system; SaMST is a recent representative emphasizing pluggable style representations, low parameter count, and fast inference [14]. These systems are directly related to our goal of efficient multi-style transfer. In contrast to representation-only style injection, our method embeds semantic cross-attention inside a latent bridge objective, allowing terminal distribution matching and kinetic regularization to explicitly shape the style-content trade-off.

Training-free and diffusion-based style transfer. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free style transfer [7], while StyleID injects style into large-scale diffusion models without training [2]. These methods can be visually strong, but diffusion sampling and semantic drift can limit their suitability for fast domain-level transfer. Our work targets a complementary efficiency regime: compact latent integration with style-id conditioning rather than iterative diffusion sampling or per-style optimization.

Table 2: Training and inference procedure for LBM.

Step	Operation
1	Sample content latent z_0 , target style id s , and target-style latent patches.
2	Construct style prior P_s and inject it through semantic cross-attention.
3	Predict endpoint velocity $v_\theta(z_0, 1, s)$ and endpoint $\hat{z}_1 = z_0 + v_\theta(z_0, 1, s)$.
4	Compute kinetic energy $\ v_\theta\ _2^2$ and terminal SWD between $\hat{z}_1$ patches and target-style patches.
5	Update θ using the weighted sum of kinetic and SWD losses.
6	At inference, integrate $z_{t+\Delta t} = z_t + v_\theta(z_t, t, s)\Delta t$ for 12 Euler steps and decode the final latent.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems [3, 16], while the Schrodinger problem connects entropic transport to stochastic path measures [13]. We use SWD as a lightweight terminal distribution matching objective. For evaluation, we combine perceptual and distributional measures including LPIPS [20], FID [8], KID [1], DISTs [5], MUSIQ [12], MANIQA [19], and style-transfer-specific ArtFID [17]. This broad metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts.

3 Method

3.1 Latent bridge formulation

Let $z_0 \in \mathbb{R}^{4 \times 32 \times 32}$ denote the VAE latent of a content image and let s denote a target style/domain identifier. The model learns a style-conditioned velocity field

$$v_\theta(z_t, t, s), \quad (1)$$

which transports content latents toward the target artistic domain. At inference time, the endpoint is obtained with explicit Euler integration:

$$z_{t+\Delta t} = z_t + v_\theta(z_t, t, s)\Delta t. \quad (2)$$

The primary evaluation uses 12 integration steps and step size 1.0. We interpret this as a bridge-inspired latent probability flow rather than a strict solver for the continuous-time Schrodinger Bridge problem. The formulation is nevertheless useful: terminal distribution matching controls where the trajectory ends, while path-energy regularization controls how far it moves from the content latent.

Algorithmic view. Training and inference can be summarized as follows. During training, a minibatch provides content latents z_0 , target style ids s , and target-style latent samples. The network predicts $v_\theta(z_0, 1, s)$, forms the endpoint $\hat{z}_1 = z_0 + v_\theta(z_0, 1, s)$, and optimizes terminal SWD plus kinetic energy. During inference, only the content latent and style id are required; the learned velocity field is integrated for a fixed number of Euler steps. This separation is important: training uses endpoint supervision for efficiency, while inference uses multiple small updates to avoid a brittle one-shot transformation.

3.2 Architecture

The implementation uses `TimeConditionedLANCETBridge(LatentAdaCUT)`. The backbone lifts the 4-channel latent to 128 channels, processes high-resolution 32×32 features, downsamples to a 16×16 body representation, applies four semantic cross-attention/body blocks, upsamples, fuses skip features, and decodes a latent velocity. The target style is represented by a learnable spatial prior

$$P_s \in \mathbb{R}^{C \times 16 \times 16}, \quad (3)$$

Table 3: Main architectural and training configuration.

Item	Value
Latent shape	$4 \times 32 \times 32$
Domains	photo, Hayao, Monet, Van Gogh, Cezanne
Base / lift channels	64 / 128
Body resolution	16×16
Body blocks	4 semantic cross-attention blocks
Decoder blocks	2
Style dimension / time dimension	160 / 256
Semantic attention temperature	0.12
Style attention temperature	0.08
Style spatial prior gain	0.35
Terminal SWD weight	20.0
Kinetic weight	1.0
SWD projections	64
SWD patch sizes	3, 5, 7, 15
Primary training length	8 epochs
Primary inference	12 Euler steps, step size 1.0

where s ranges over {photo, Hayao, Monet, Van Gogh, Cezanne}. This style prior is a domain-level prototype rather than a per-reference image representation. It enables fast inference but also limits raw reference-specific style capacity.

Semantic cross-attention injects the style prior into content features:

$$Q = f_c(x), \quad K, V = f_s(P_s), \quad A = \text{softmax}(QK^\top / \tau), \quad \hat{x} = AV. \quad (4)$$

The primary model uses semantic attention temperature 0.12, style attention temperature 0.08, and style attention sharpen scale 2.5. Skip fusion uses an additive projected route to preserve spatial content structure.

3.3 Training objective

Training uses an OMF-style endpoint approximation. For a content latent z_0 and target style s , the model predicts a velocity at the endpoint training time and forms

$$\hat{z}_1 = z_0 + v_\theta(z_0, 1, s). \quad (5)$$

The loss combines kinetic regularization and terminal SWD:

$$\mathcal{L} = \lambda_{\text{kin}} \mathcal{L}_{\text{kin}} + \lambda_{\text{swd}} \mathcal{L}_{\text{swd}}. \quad (6)$$

The kinetic term is

$$\mathcal{L}_{\text{kin}} = \mathbb{E} \|v_\theta(z_0, 1, s)\|_2^2, \quad (7)$$

which penalizes excessive latent displacement. The terminal SWD term compares generated endpoint patches with target-style patches over 64 sliced projections and patch sizes 3, 5, 7, and 15. The primary configuration uses $\lambda_{\text{kin}} = 1.0$ and $\lambda_{\text{swd}} = 20.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This separation between endpoint style distribution matching and path-energy regularization is the central mechanism behind the observed style-content Pareto behavior.

Why SWD rather than paired reconstruction. The target artistic domain is represented by unpaired examples, so there is no correct pixel-level target for a content image. A paired reconstruction loss would incorrectly force spatial alignment with an unrelated artwork. SWD instead compares projected patch distributions. In one dimension, the Wasserstein distance can be computed by sorting projected samples; averaging over many projections yields an efficient approximation to high-dimensional distribution matching.

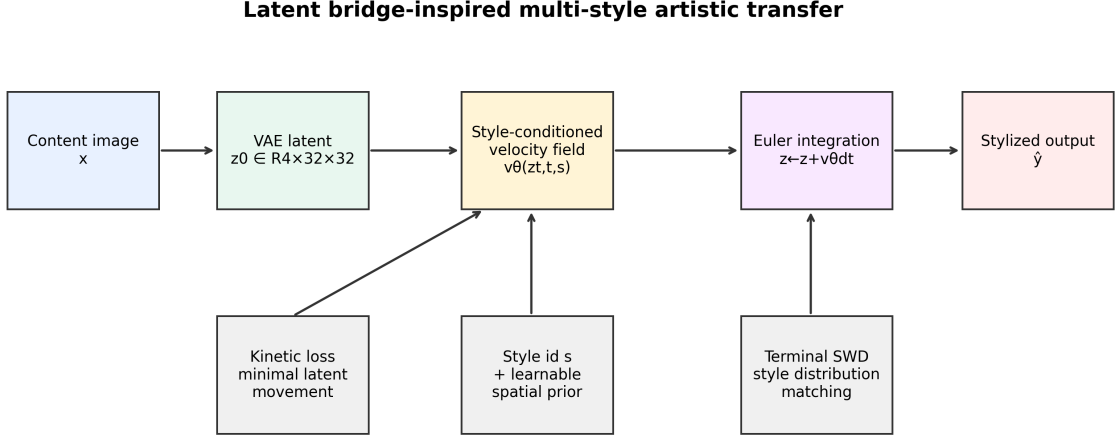


Figure 1: Overview of the proposed latent bridge-inspired style transfer framework. A content image is encoded into a compact latent, transported by a style-conditioned velocity field, constrained by terminal SWD and kinetic regularization, and decoded into a stylized output.

In our implementation, multi-scale patch sizes make the terminal loss sensitive to both fine texture and broader style statistics.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^M$, the empirical terminal objective is

$$\mathcal{L}_{\text{swd}} = \sum_{r \in \mathcal{R}} \frac{1}{M} \sum_{m=1}^M \|\text{sort}(\mathcal{P}_r(\hat{z}_1)\theta_m) - \text{sort}(\mathcal{P}_r(z_s)\theta_m)\|_2^2. \quad (8)$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics, not whether it matches any particular artwork spatially.

Semantic-guided projections. The implementation also supports style-aware projection selection. Rather than relying only on random directions, semantic keys from the style/content routing module identify feature directions with high style response, and SWD is computed along these directions. This gives the terminal loss a closer connection to the style transport module: the same semantic structure used to inject style also influences which endpoint statistics are matched. This is a practical approximation rather than an exact optimal-transport coupling, but it makes the style objective less blind than uniform random projection alone.

Bounded theoretical claim. The method is not an exact Schrodinger Bridge solver. It does not estimate forward and backward stochastic scores, nor does it optimize the full entropic dual objective. The bridge perspective is used to design a tractable training principle: a terminal distribution constraint plus a kinetic path penalty. This bounded interpretation is important for rigor and is also reflected empirically. Removing the terminal constraint weakens style, while removing the path penalty destroys content.

4 Experiments

4.1 Protocol and metrics

We evaluate under a strict 750-output protocol: five source domains times five target domains times 30 source images. The five domains are photo, Hayao, Monet, Van Gogh, and Cezanne; Ukiyo-e is not used in the

Table 4: Baseline completion status in the current local study.

Method	Status	Role
AdaIN v32k/vgg19	complete strict-750	classical fast AST
SaMST	complete strict-750	multi-style efficiency baseline
StyleID	complete strict-750	diffusion-style baseline
S2WAT	complete strict-750	transformer baseline
StyTr2	smoke-tested	future complete run
CAST	smoke-tested	future complete run
AesFA	timing probe only	future complete run
AesPA-Net	prepared only	future complete run

Table 5: Strict-750 comparison. Higher is better for CLIP-S, CLIP-C, EC, MUSIQ, and MANIQA. Lower is better for LPIPS, DISTS, HF-KID, and KID.

Method	CLIP-S	CLIP-C	LPIPS	EC	MUSIQ	MANIQA	DISTS	HF-KID	KID
Ours K1 epoch 7	0.7161	0.8086	0.4514	0.3928	49.2059	0.4057	0.2477	4.1694	0.0524
Ours K1 epoch 8	0.7167	0.7977	0.4615	0.3859	—	—	—	—	0.0554
SaMST strict	0.7194	0.8193	0.4664	0.3839	36.0950	0.3139	0.2943	6.7598	0.0489
StyleID strict	0.7597	0.5519	0.7497	0.1902	—	—	—	—	—
S2WAT strict	0.7139	0.7465	0.5263	0.3382	36.5256	0.1754	0.2942	12.6623	0.0567
AdaIN v32k	0.7130	0.6990	0.6298	0.2639	—	—	—	—	—
AdaIN vgg19	0.6930	0.5991	0.6870	0.2169	—	—	—	—	—

present protocol. We report CLIP-style, CLIP-content, LPIPS-content, and the composite score

$$EC = \text{CLIP-style} \cdot (1 - \text{LPIPS}). \quad (9)$$

For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, and shallow Gram micro statistics. Complete strict-750 local outputs are available for Ours, SaMST, StyleID, S2WAT, and AdaIN variants.

Metric taxonomy. We use four metric groups. Raw semantic metrics, CLIP-style and CLIP-content, measure whether an output is recognized as stylistically aligned and semantically related to the input. Perceptual content metrics, LPIPS and DISTS-content, measure visual structure preservation. Distributional style metrics, including KID, Gram statistics, and high-frequency patch KID, evaluate whether generated texture statistics resemble target-style images. No-reference image-quality metrics, MUSIQ and MANIQA, help detect dirty or over-processed outputs. EC is used only as an internal Pareto diagnostic for sweep selection; all claims are checked against the individual metrics.

Table 4 is included to avoid overstating the comparison set. The quantitative claims in the main paper are restricted to methods with complete strict-750 outputs. Smoke-tested or prepared methods are treated as future baselines rather than as matched-protocol comparisons.

4.2 Main comparison

Table 5 shows that Ours epoch 7 is close to SaMST in raw CLIP-style while improving LPIPS and EC. StyleID obtains the highest CLIP-style, but its CLIP-content of 0.5519 and LPIPS of 0.7497 indicate severe semantic drift. AdaIN variants are substantially weaker in content preservation. S2WAT reaches a comparable style score but has worse LPIPS and much worse HF-Patch-KID.

4.3 Artifact-sensitive comparison to SaMST

SaMST is a strong baseline under raw style and structure metrics, but the side-by-side grids in Figure 3 reveal muddy and grain-like artifacts. We therefore examine artifact-sensitive diagnostics in Table 6. Ours

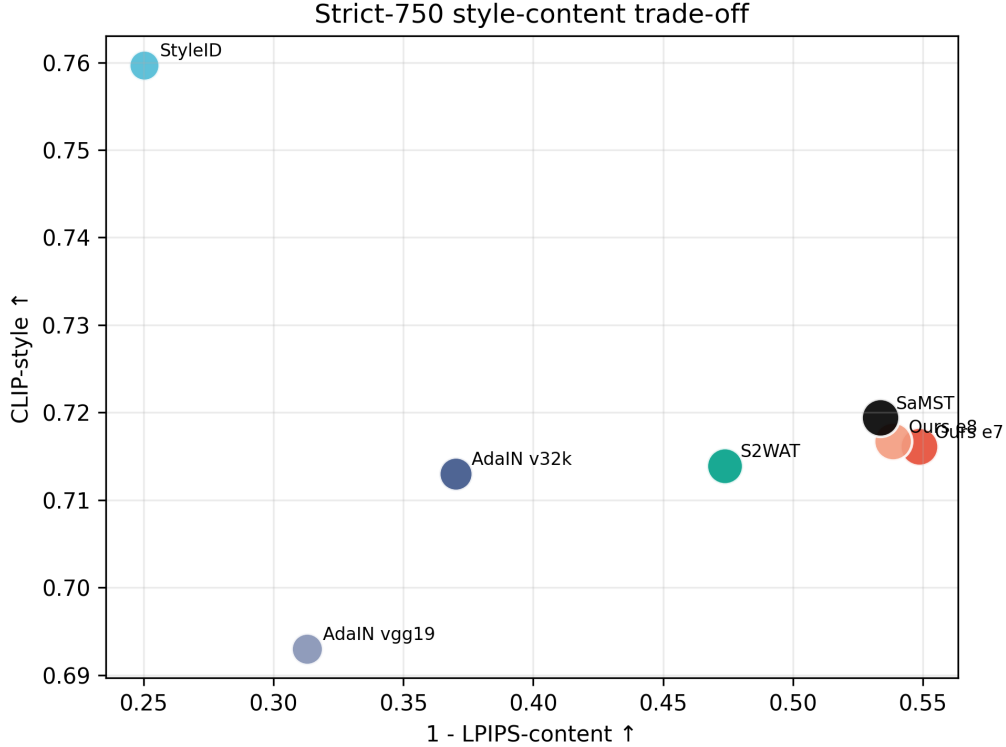


Figure 2: Strict-750 style-content trade-off. Ours is close to SaMST in raw CLIP-style while retaining a better LPIPS-based trade-off than several baselines; StyleID reaches high style but collapses content.

improves MUSIQ by 13.1109, MANIQA by 0.0918, DISTS-content by 0.0466, HF-Patch-KID by 2.5904, and FFT slope error by 0.5063. These results support the interpretation that SaMST preserves low-frequency structure but produces less realistic local texture statistics.

This diagnostic block addresses a specific evaluation pitfall. Metrics such as CLIP-content, SSIM, and edge overlap tend to reward low-frequency structure preservation. A grainy output can therefore score well if object boundaries remain visible, even when the surface texture is visually dirty. The added diagnostics target a different failure mode: whether local high-frequency patches, no-reference perceptual quality, and frequency slopes resemble real artistic texture rather than spatially incoherent dithering. These metrics should not replace human evaluation, but they provide objective evidence for the artifact mode observed in visual inspection.

4.4 Destructive ablations

The ablation results in Table 7 expose the mechanism of the model. Removing terminal SWD gives excellent content preservation but weak style, showing that terminal distribution matching is the primary style driver. Removing kinetic regularization preserves style but collapses content, confirming that the path-energy penalty is essential. Strong color loss damages both content and distributional quality, so naive color matching is not used in the mainline.

4.5 Weight sweep and theory switches

The 40-run category-weight sweep trained 20 sampling recipes with $K \in \{1, 2\}$ for eight epochs each, evaluating every epoch for 320 total checkpoints. The best EC result was K2 balanced default at epoch 3 with CLIP-style 0.6980, CLIP-content 0.8727, LPIPS 0.3777, and EC 0.4343. The best raw-style result in the sweep was K1 balanced default at epoch 8 with CLIP-style 0.7161, CLIP-content 0.7984, LPIPS 0.4605,

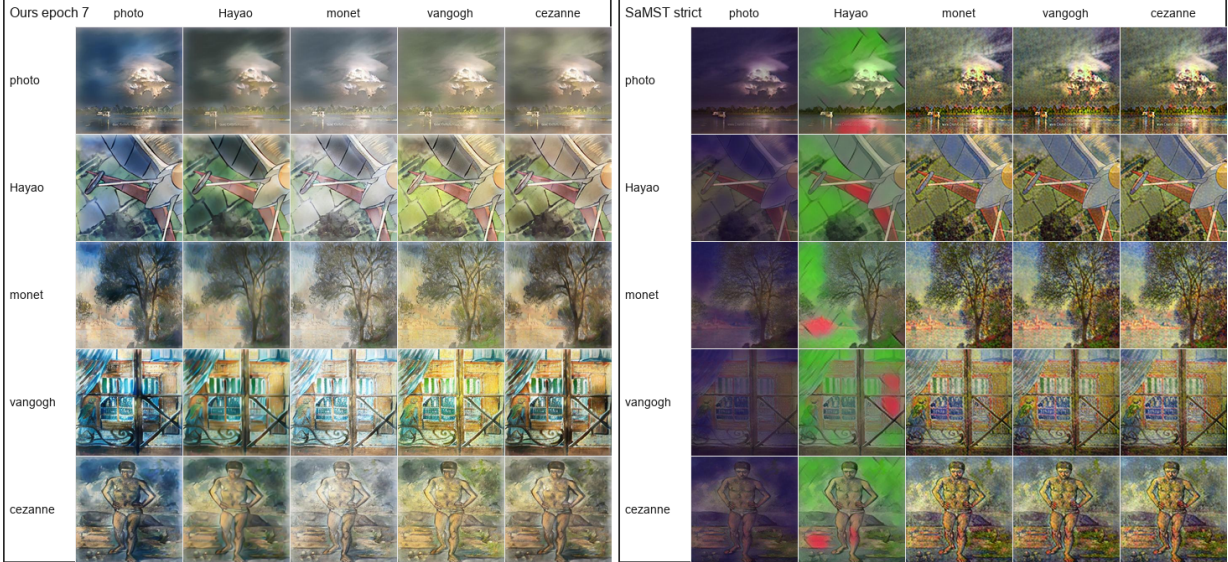


Figure 3: Qualitative 5-by-5 source-target grids for Ours epoch 7 and SaMST strict. Rows indicate source domains and columns indicate target domains. SaMST preserves coarse structure but often produces a muddy, grain-like surface texture; Ours is less aggressive in raw style but visually cleaner and less dithering-like.

Table 6: Artifact-sensitive comparison between Ours and SaMST.

Metric	Ours e7	SaMST
MUSIQ $\uparrow$	49.2059	36.0950
MANIQA $\uparrow$	0.4057	0.3139
DISTS-content $\downarrow$	0.2477	0.2943
HF-Patch-KID $\downarrow$	4.1694	6.7598
FFT slope error $\downarrow$	0.5473	1.0536
Gram micro $\downarrow$	0.0798	0.0947

and EC 0.3863. The sweep showed that more complex category weighting was not consistently better than balanced sampling.

We also evaluated optional theory switches. Sinkhorn-normalized semantic routing and entropy-gated kinetic regularization improved EC/content but reduced raw CLIP-style. The best theory-switch EC was entropy-gated kinetic with weight 5.0 at epoch 1, giving CLIP-style 0.6916, CLIP-content 0.8804, LPIPS 0.3684, and EC 0.4368. Color transport increased raw style slightly but harmed LPIPS and content, and was therefore not selected as the mainline.

4.6 Resource use and reproducibility

The experiments were run from a reproducible workspace with saved configs, checkpoints, strict-750 outputs, metrics, and logs. The primary configuration is stored in the K1-SWD20 run directory. Destructive ablations, the 40-run weight sweep, and theory-switch validation are stored in separate experiment directories with their own configs and training logs. The D0 full destructive-control run recorded approximately 290.65 seconds for the measured seven-epoch setting. Additional timing reports are stored with the related-work evaluation outputs, but we conservatively treat timing as a supporting engineering result rather than a primary quality claim until all baselines are measured under identical hardware conditions.

All reported strict-750 evaluations use the same domain protocol and local CLIP cache. We also evaluated every epoch in the 40-run sweep, yielding 320 checkpoint evaluations. This matters because final-epoch selection alone can hide style-content trade-offs: K2 balanced default achieved the best EC at epoch 3,

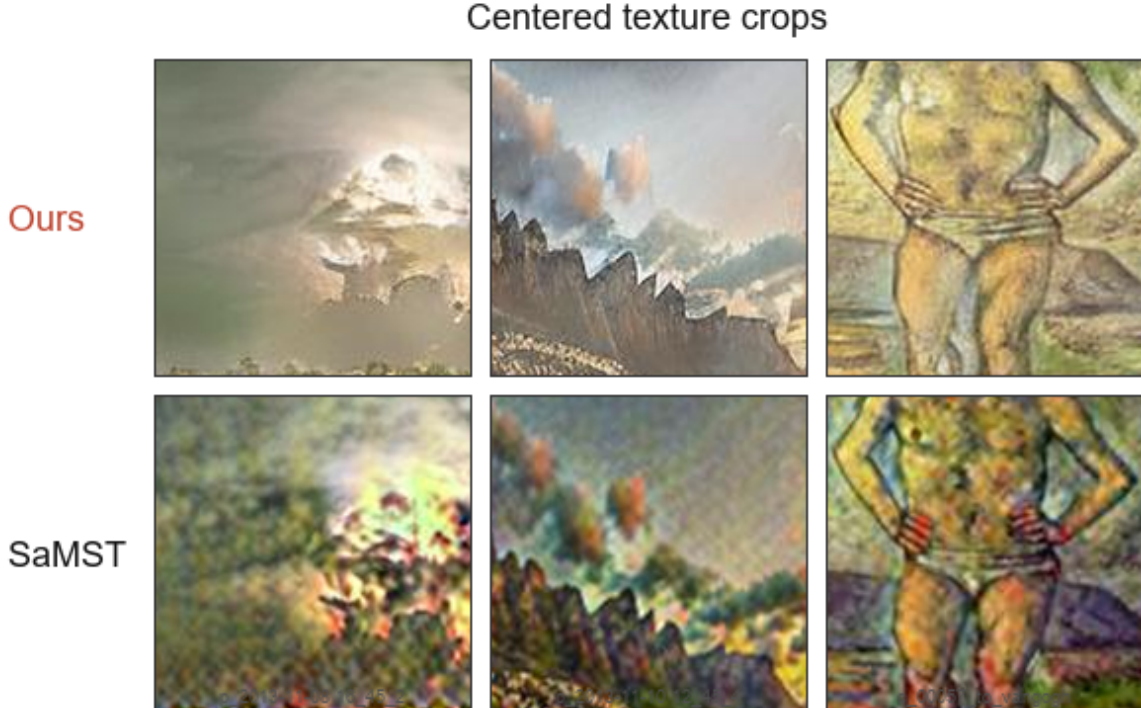


Figure 4: Centered texture crops from matched Ours and SaMST outputs. The crops isolate the surface-texture failure mode: SaMST often retains coarse layout but introduces grain-like, spatially incoherent texture, whereas Ours is visually cleaner.

Table 7: Destructive ablations under the strict-750 protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o terminal SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

whereas K1 balanced default achieved the best raw style at epoch 8.

5 Discussion and Limitations

The experiments support a specific and conservative conclusion. The proposed method does not universally dominate all baselines on raw style: StyleID has higher CLIP-style, and SaMST is slightly higher than Ours epoch 7 on raw CLIP-style and CLIP-content. However, StyleID suffers content collapse, while SaMST exhibits grain-like artifacts that are poorly captured by standard structure metrics. Ours provides a cleaner style-content trade-off and stronger artifact-sensitive perceptual quality under the local strict-750 protocol.

The current study has several limitations. Complete strict-750 runs are not yet available for CAST, AesFA, AesPA-Net, and StyTr2, so they should not be over-discussed as fully matched baselines. User-study data has not yet been collected, which is important for validating perceptual cleanliness and artifact preference. Finally, the style representation is domain-level rather than reference-image-specific; this makes inference efficient but limits the ability to reproduce unique single-image brushstroke patterns. Future work should evaluate controlled integration horizons, semantic paint gains, K schedules, richer domain prototype priors, high-resolution generalization, and human preference studies.

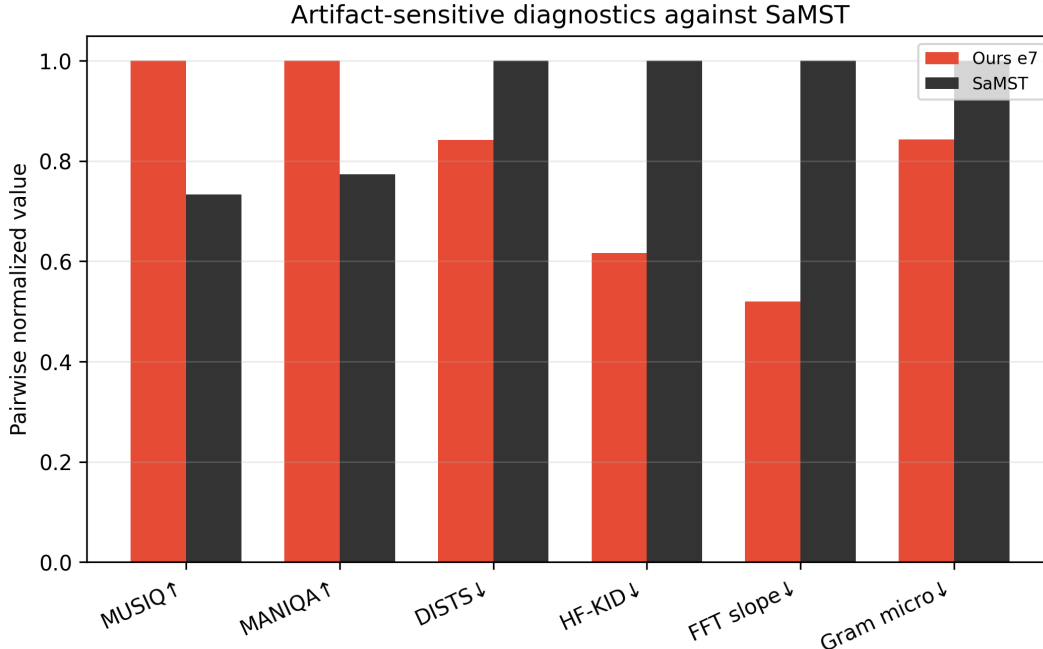


Figure 5: Artifact-sensitive diagnostics for Ours and SaMST. Mixed-scale metrics are pairwise normalized for readability; the raw numeric values are reported in the experimental table.

Table 8: Measured or estimated runtime from local logs. Values are not yet a fully controlled hardware benchmark, so they are used as engineering evidence rather than the primary quality claim.

Method	Train sec	Infer sec/img	Note
Ours	309.902	0.1139	train to epoch 7
SaMST	2030.610	0.0531	train extrapolated
StyleID	—	4.0218	estimated full 750
AdaIN v32k	9220.393	0.0124	full local run
AdaIN vgg19	262.780	0.0121	full local run

6 Conclusion

We presented a latent bridge-inspired framework for fast multi-style artistic transfer. By combining terminal SWD style distribution matching with kinetic path regularization, the method separates the forces that drive style from those that preserve content. A lightweight style spatial prior and semantic cross-attention make this formulation practical for efficient domain-level transfer. Under a strict 750-output evaluation, the method reaches a competitive raw style score while improving LPIPS and artifact-sensitive perceptual diagnostics over SaMST and preserving content substantially better than StyleID. The results indicate that latent bridge matching is a promising foundation for fast, clean, and reproducible artistic style transfer.

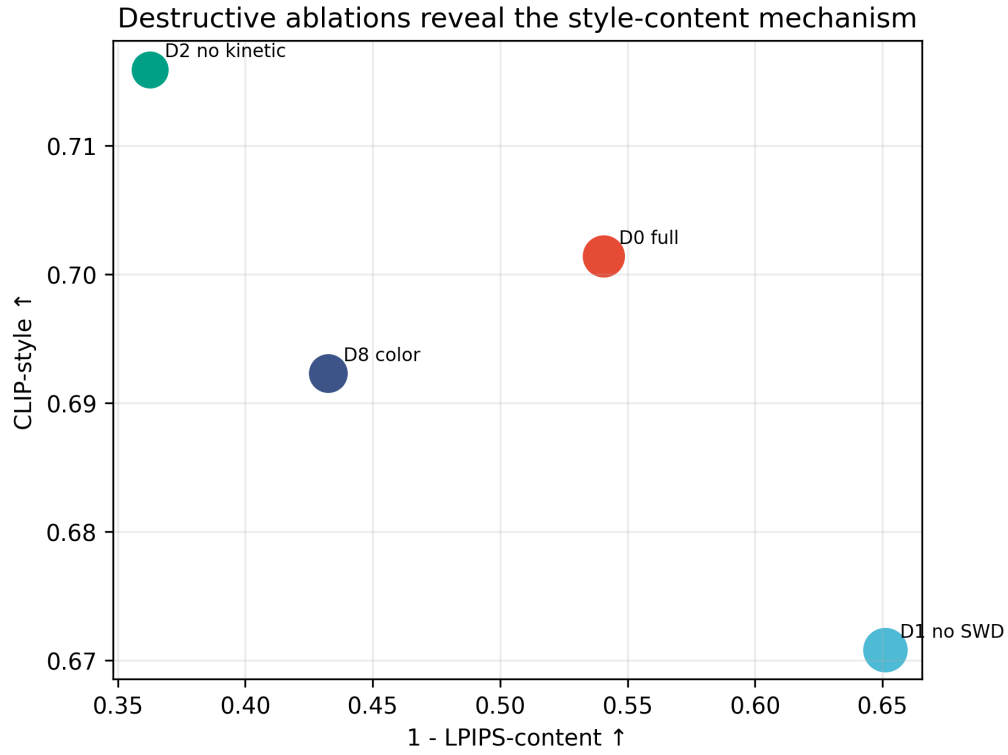


Figure 6: Destructive ablations. Removing terminal SWD improves content but weakens style, while removing kinetic regularization increases style at the cost of severe content degradation.

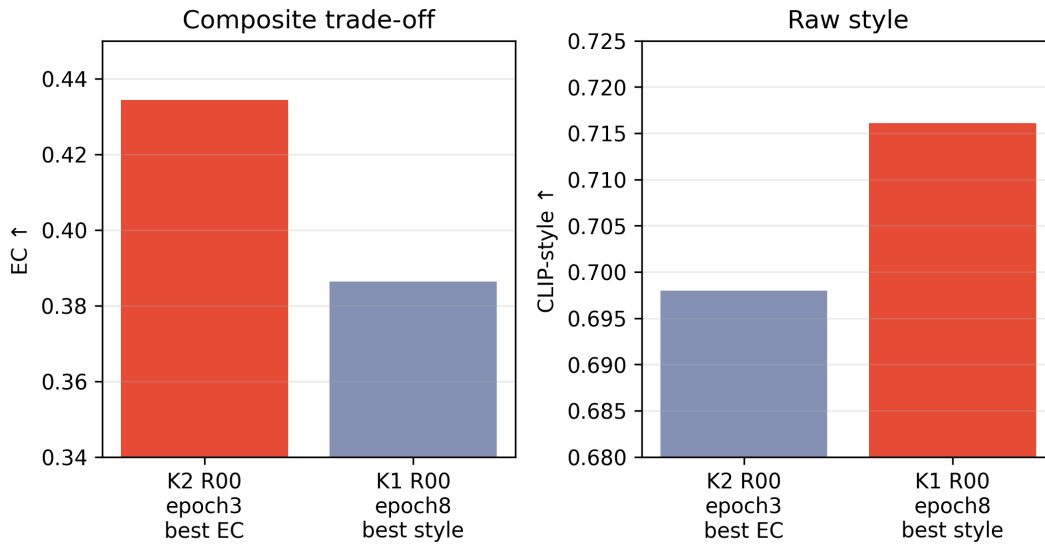


Figure 7: Summary of the 40-run category-weight sweep. K2 balanced default gives the best EC, whereas K1 balanced default gives the best raw CLIP-style among evaluated checkpoints.

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